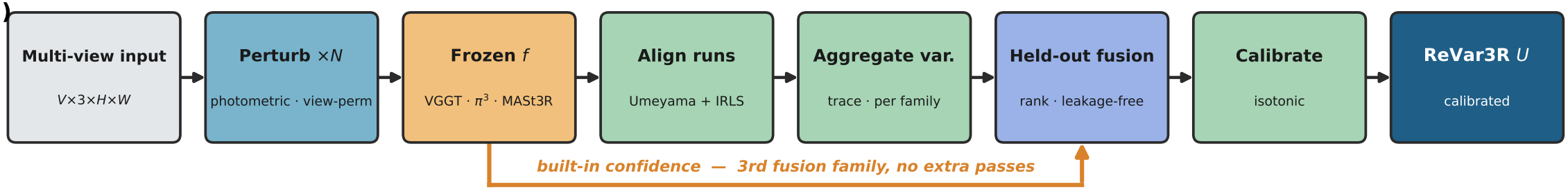
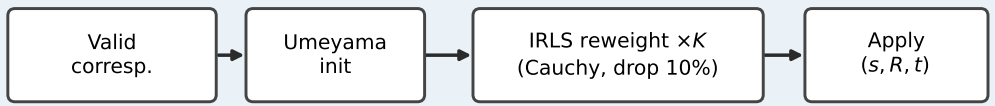
ReVar3R — training-free, calibrated uncertainty for frozen feed-forward 3D models

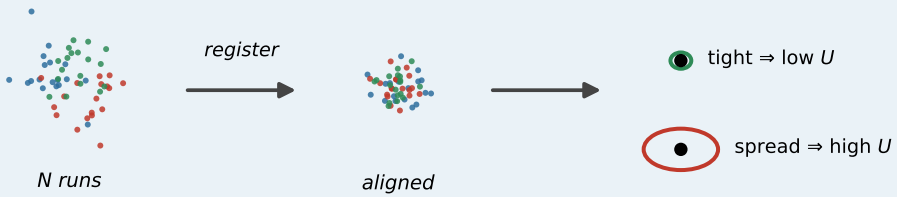
A · Input | B · Perturb & predict (frozen, ×N) | C · Per-point uncertainty | D · Fuse & calibrate | E · Output

(a)



(b) Align → variance · detail of stage C





$$(s, R, t) = \operatorname{argmin} \sum_i \rho(\|sR\, p_i^{(n)} + t - \bar{p}_i\|), \quad \rho = \text{Cauchy}$$

$$\tilde{U}_i = \frac{1}{N} \sum_{n=1}^{N} \|\hat{p}_i^{(n)} - \bar{p}_i\|_2^2$$

naive variance (no alignment) ≈ random — alignment-before-variance is load-bearing

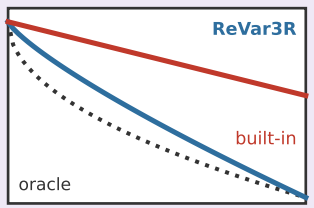
(c) Fuse · calibrate · exploit · detail of stages D–E

$$U = \sum_k w_k r_k, \quad r_k = \operatorname{rank}(\tilde{U}^k), \quad w_k \propto \text{AUSE}_k^{-2}$$

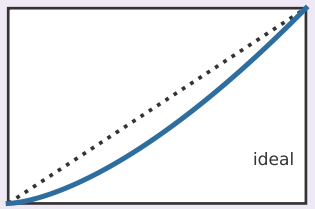
$k \in \{\text{photometric, view-perm, built-in}\}$, weights fit on a held-out split (leakage-free)

calibrate: $\hat{U} = \text{isotonic}(U)$ — ranking & fusion are calibration-free (need no GT)

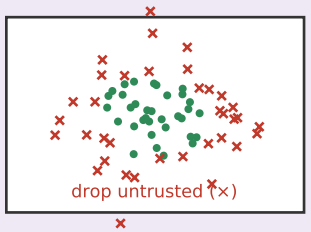
Sparsification → AUSE ↓



Reliability → ECE ↓



MV fusion → Acc/Comp ↓



ReVar3R tracks the oracle exactly where the model's built-in confidence fails.